

AMARNATH S. PATEL

Orlando, FL | amarnath.patel@ucf.edu | apatel.co | github.com/jeebuscrossaint

Systems and scientific software — C++23/CUDA, Rust, Python. Hardware to GPU kernels to shipped desktop apps.

EDUCATION

University of Central Florida

B.S. Physics (Optics & Lasers) and Mathematics, Computer Science minor

Undergraduate Student 4.00 GPA

August 2025 – Expected May 2029

- **Computer science:** Data Structures & Algorithms, Computer Architecture, Computer Logic Design, Object-Oriented Programming, Discrete Structures, C Programming, Deep Learning
- **Mathematics:** Linear Algebra (proof-based, in progress), Complex Analysis, Partial Differential Equations (in progress), Applied Linear Algebra, Ordinary Differential Equations, Honors Calculus III

Florida Atlantic University

111 credit hours completed between ages 14 and 18, alongside high school

University coursework via FAU High School, ages 14–18 3.66 GPA

August 2021 – May 2025

EXPERIENCE

Undergraduate Researcher, Instrumentation Software — UCF Astrophotonics Lab, CREOL

Aug 2025 – Present

- Own the lab's [acquisition software](#): a Python stack that drives four instruments as one system — tunable IR laser, InGaAs camera, fiber switch and motorized polarization controller — across three transports (GPIB/VISA, RS-232, GigE Vision), so a full all-port × C-band measurement sweep runs unattended. Days of manual bench operation became a single hands-off run.
- Closed-loop control and fault handling in the acquisition path: polarization optimized in-loop against a live fringe-contrast objective, saturated frames detected and rejected as they arrive, so an overnight run does not silently produce unusable data.
- Built the numerical pipeline that turns raw frames into results — FFT sideband isolation and demodulation, Butterworth filtering, and a joint nonlinear optimization over three coupled parameters (NumPy/SciPy) — validated at ≈98% reconstruction fidelity against simulated ground truth.
- Contribute mount control and pointing automation (INDI) to [PolyOculus](#), an instrument program coordinating a networked array of 8 telescopes.

Undergraduate Research Assistant — UCF Physics Department

Mar 2026 – Present

Physics Education Research — Dr. Zhongzhou Chen (NSF Award 2421299; Gates Foundation INV-076932)

- Built [ESTELA](#) (Rust, Tauri 2), a desktop application that generates multi-version isomorphic exams from a structured bank of 615 problems across 29 banks, 13 topic areas and 11 question types, with auto-generated answer keys and export to four formats. Shipped to external adopters outside the university following a faculty workshop.
- Built an agentic generation–evaluation pipeline over the Azure OpenAI batch API that drafts candidate problems and validates them against instructor criteria before a human ever reviews them.

Programmer, Coding Subteam — Advanced Experimental Vehicles, Florida Atlantic University

Nov 2023 – May 2025

- Owned the Unix and systems half of a four-person software team building live telemetry for a four-passenger solar car: Arch Linux ARM on a Raspberry Pi 5, a Hyprland compositor session driving the in-car touchscreen, gpsd and USB-serial plumbing to the GPS and Thunderstruck BMS, and the WireGuard tunnel that made worldwide remote monitoring possible at all.
- The dashboard those services fed (team stack: Nuxt.js, Tailwind, Electron and Three.js over an Express + WebSocket backend) ran in the car and streamed to the pit crew and to viewers off-site simultaneously, through four days of racing. 2nd in the Cruiser Division; Lockheed Martin Award for engineering excellence.

SELECTED SOFTWARE

CELERIS — electromagnetic solver and metalens design pipeline. C++23, CUDA, Python, CMake

2026

- Wrote a Maxwell solver from scratch rather than wrapping an existing kernel, then built the design and analysis tooling on top: parameter sweeps, gradient-based (Adam) inverse design, Monte-Carlo tolerance analysis, and fabrication-ready GDSII export. Python bindings and a native Qt desktop GUI.
- Correctness is enforced, not assumed: a validation suite checks every layer against closed-form solutions, an independent method, or energy conservation (agreement from 1e-6 to 1e-15), and cross-checks the whole solver against two third-party implementations to ≈1e-7.
- Implemented Rayleigh–Sommerfeld propagation as a CUDA kernel and benchmarked it against the project's own optimized 16-core CPU path: 4.8–5.8× at 92k elements, agreeing to machine precision, with automatic CPU fallback. Diagnosed the kernel as memory-bound rather than compute-bound from the way the speedup declined as the grid grew. *Sole-author manuscript in preparation.*

Systems and tooling — C, C++, Rust, Go

2023 – Present

- [hush](#), an interactive shell in C; [limebar](#), a featherweight Wayland status bar in C; [constrict](#), a build system in Go written to replace make; [coat](#), a universal theme applier in Rust; [hazmat](#), a small password manager in C99. Tools I use daily on my own machine, not demos.
- [slashbinslash.sh](#), temporary file sharing and storage; [apatel.co](#), a hand-written static site with no framework, no build step and no third-party requests at runtime; UniUtils, a course-schedule generator and classroom finder — 1st place, Night Hacks hackathon.

TECHNICAL SKILLS

Languages: C, C++ (C++23), Rust, Python, Go, JavaScript/TypeScript, CUDA, VHDL, Shell (Fish, Bash)

Frameworks and libraries: Qt/PySide6, Tauri 2, Nuxt/Vue, Express, Electron, Three.js, NumPy, SciPy

Systems and tools: Linux (Arch, daily driver), Git, CMake, Docker, WireGuard, Wayland, embedded Linux on Raspberry Pi, GPU/CUDA profiling, GPIB/VISA, RS-232, GigE Vision, LaTeX/Typst

HONORS

Florida Bright Futures — Florida Academic Scholars (highest tier; 100% tuition)

2025

Lockheed Martin Award, “Highest Level of Engineering Excellence,” AEV Competition

2024

1st Place, Night Hacks Hackathon — UniUtils

2023